

PROBLEM STATEMENT

Core Pain Points & Root Cause Analysis



Lack of Personalized AI-Driven Maternal & Fetal Insights

- ◆ **Issue:** Generic pregnancy advice overlooks individualized risks (e.g., gestational diabetes, fetal distress).
- ◆ **Root Cause:** No AI-driven analytics to correlate maternal vitals (B P , glucose, HR V) with fetal development trends for early risk detection.



Low Expecting Mother Engagement & Motivation

- ◆ **Issue:** Expecting mothers often skip vital prenatal routines, leading to undetected health risks.
- ◆ **Root Cause:** Lack of habit-reinforcing AI nudges, gamified milestone tracking, voice-assisted reminders, and stress-relief features for consistent health adherence.



0 3 . Fragmented Health Data & Poor Integration

- ◆ **Issue:** Disjointed tracking—smart bands monitor vitals, prenatal checkups record fetal health, lifestyle apps track nutrition, but data isn't synchronized for insights.
- ◆ **Root Cause:** Absence of real-time IoT integration combining wearable sensors, E HR s, prenatal screening reports, and AI-driven analytics in a unified dashboard for doctors & users.



0 4 . Privacy & Security Concerns

- ◆ **Issue:** Expecting mothers fear unauthorized access to their sensitive health data (e.g., pregnancy complications, fetal anomalies).
- ◆ **Root Cause:** Weak AES -256 encryption, no blockchain-based medical record integrity, poor user-controlled access and lack of regulatory compliance (GDPR , HIPAA).



0 5 . Limited Accessibility Across Devices & Ecosystem

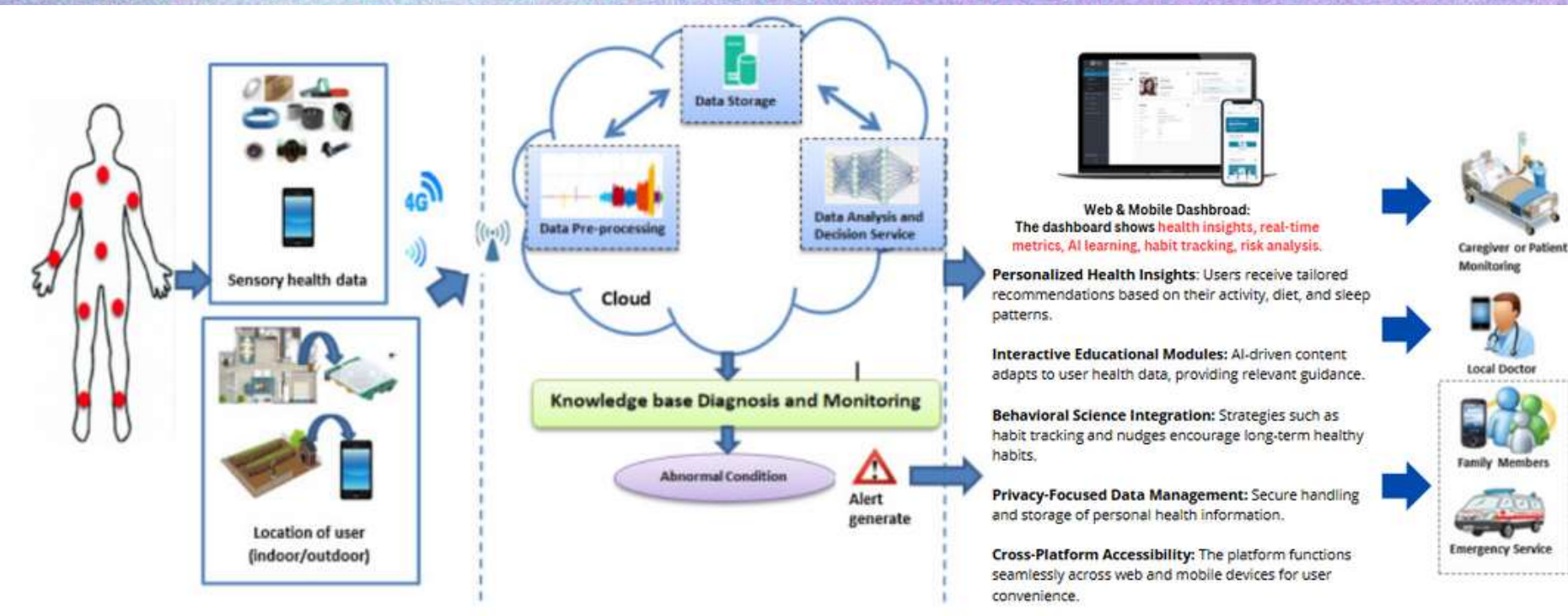
- ◆ **Issue:** Health insights are not accessible across platforms.
- ◆ **Root Cause:** No seamless integration between mobile, web, and wearable technology.

Relevancy to the Chosen Area of Focus :

- ◆ The solution is focused on **maternal and fetal health monitoring**.
- ◆ Complications like **preeclampsia, gestational diabetes, and fetal distress** often go undetected early
- ◆ Limited real-time monitoring.
- ◆ Lack of **predictive risk assessment**.
- ◆ **Poor communication between doctors and patients**.
- ◆ This **issue is critical in rural and underserved areas** where consistent medical checkups are not available.

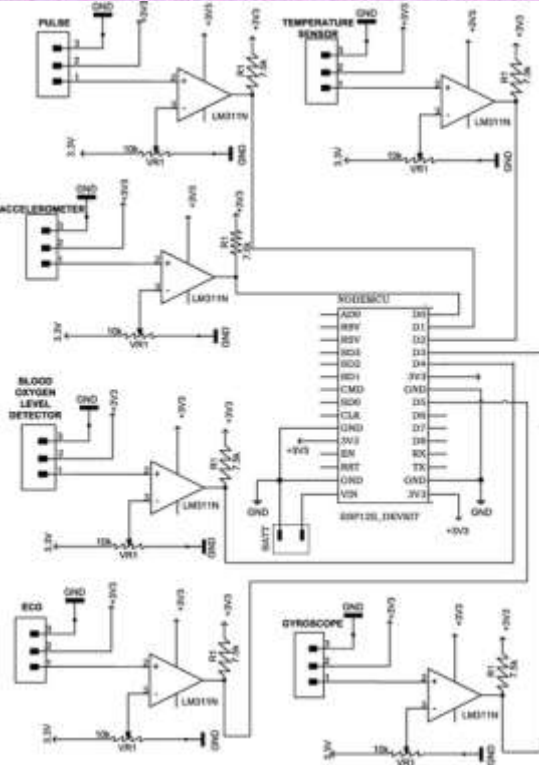
PROPOSED SOLUTION

Workflow:



SmartHealth Band

The aim is to develop a smart HealthBand that continuously monitors health parameters and transmits data to the cloud. The server will process and analyze the data, providing tailored health recommendations based on activity, diet, and sleep patterns. Additionally, AI-driven educational modules will adapt to user health data, offering personalized guidance for improved well-being.



AI-POWERED PERSONALIZATION

(Solves Lack of Personalization in Health Education)

- ◆ **Adaptive AI models** analyze user data (activity, diet, sleep) to generate **personalized health recommendations**.
- ◆ Dynamic learning modules adjust in real-time based on user habits and goals.

GAMIFIED ENGAGEMENT & HABIT REINFORCEMENT

(Boosts User Motivation & Consistency)

- ◆ **Rewards, challenges, and social leaderboards** foster engagement.
- ◆ **AI-driven nudges** reinforce positive health behaviors over time.

UNIFIED HEALTH DATA ECOSYSTEM

(Eliminates Fragmented Data Silos)

- ◆ **Seamless integration with wearables, medical records, and lifestyle apps.**
- ◆ **Real-time AI analytics** transform raw data into **actionable insights**.

PRIVACY-FIRST & SECURE DATA HANDLING

(Overcomes Security Concerns)

- ◆ **End-to-end encryption, anonymization, and strict user controls.**
- ◆ **HIPAA/GDPR-compliant architecture** ensures full transparency and security.

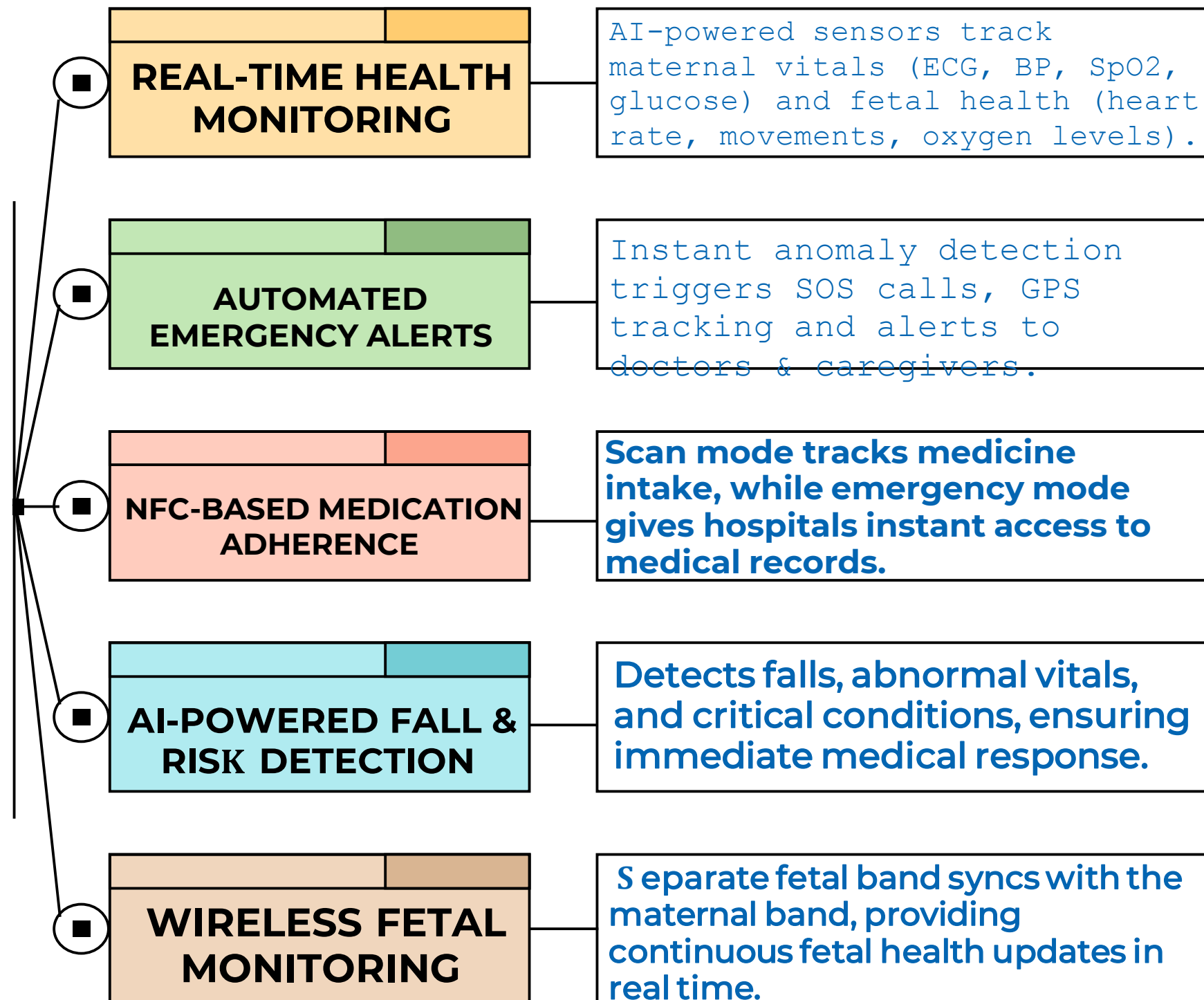
CROSS-PLATFORM, ALWAYS-ACCESSIBLE INSIGHTS

(Solves Limited Accessibility Issues)

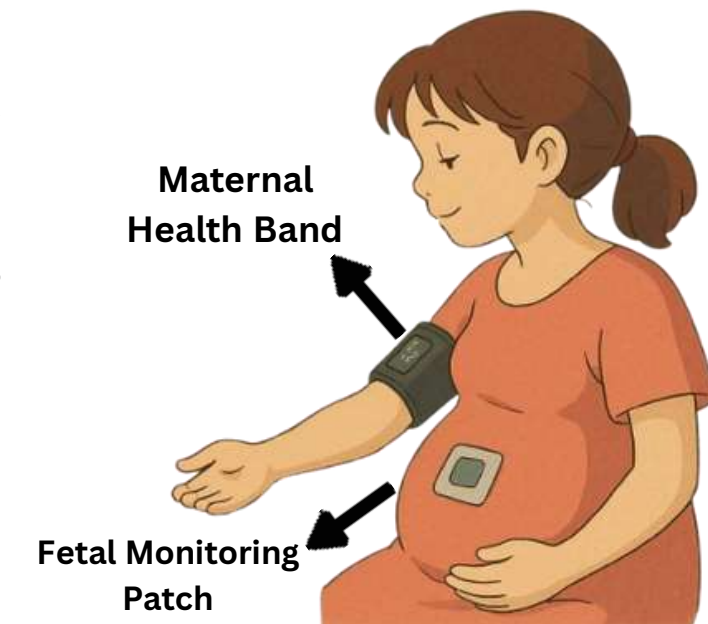
- ◆ **Unified experience across mobile, web, and wearables** with **cloud-sync**.
- ◆ **AI chatbots & voice assistants** deliver **instant health insights** on demand.

PROPOSED SOLUTION

Innovative Solutions



The aim is to develop a smart health band system that continuously monitors maternal and fetal health parameters and transmits data to the cloud. The server will process and analyze this data, providing personalized health insights for both mother and baby. Additionally, AI-driven predictive analytics will assess risks, while adaptive educational modules will offer tailored pregnancy guidance based on vital signs, activity, and overall health trends.



How It Improves & Prevents Maternal & Fetal Health Risks

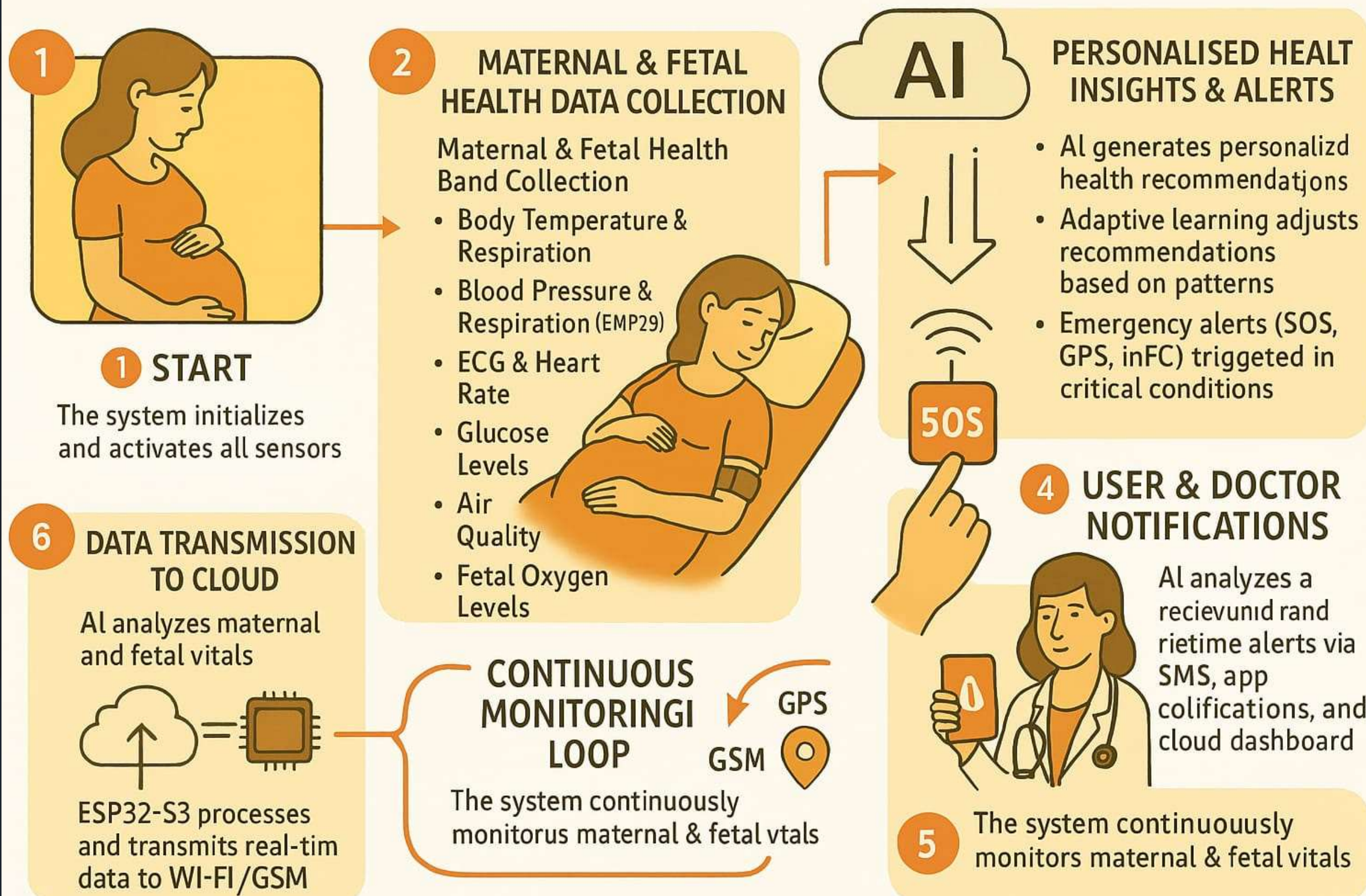
Improves Pregnancy Care :

- ◆ AI-powered monitoring ensures real-time tracking of maternal and fetal vitals.
- ◆ Personalized health insights help mothers make informed decisions.
- ◆ Seamless doctor connectivity allows timely medical intervention.

Prevents Complications :

- ◆ Early risk detection prevents preeclampsia, gestational diabetes & fetal distress.
- ◆ SOS & GPS alerts ensure instant medical help in emergencies.
- ◆ NFC tracking prevents missed medications, reducing health risks.

FLOW CHART



The flowchart represents an AI-powered maternal and fetal health monitoring system that continuously tracks vital health parameters. The process begins with the system initialization, activating sensors that measure body temperature, respiration, blood pressure, ECG, blood oxygen levels, fetal kicks, and air quality. AI analyzes this data to generate personalized health recommendations and detect anomalies. If an emergency is detected, the system triggers alerts via SOS, GPS, and NFC to ensure immediate attention.

Once an alert is generated, real-time notifications are sent to doctors and caregivers through SMS, mobile app notifications, and cloud dashboards. The system continuously monitors health data, looping back for ongoing tracking to ensure proactive healthcare. All processed data is securely transmitted to the cloud via Wi-Fi or GSM, allowing remote access for medical professionals and enabling timely interventions when necessary.



TECH STACK

1. Data Collection (Edge Layer)

Wearable sensors (Heart rate, Sleep, Activity, Diet).

IoT Gateway (Raspberry Pi) transmits data via Wi-Fi/Bluetooth/LTE.

2. Cloud Storage & Transmission

MQTT/HTTPS for secure data transfer.

NoSQL (MongoDB) & SQL (MySQL) for data.

3. AI-ML Processing & Analytics

Real-time data cleaning (Kafka & Spark).

AI models (TensorFlow & PyTorch) for behavior & risk analysis.

Personalized health recommendations via smart habits tracking.

4. User Interaction (Apps & API Layer)

Mobile & Web Apps (React, Flutter) for dashboards & tracking.

API Gateway (FastAPI & Node.js) for seamless data access.

Gamification (Streaks, Rewards, Notifications) for engagement.

5. Security & Compliance

OAuth 2.0 for authentication.

GDPR-compliant encryption (AES-256) for privacy.

Data anonymization & federated learning for secure AI processing.

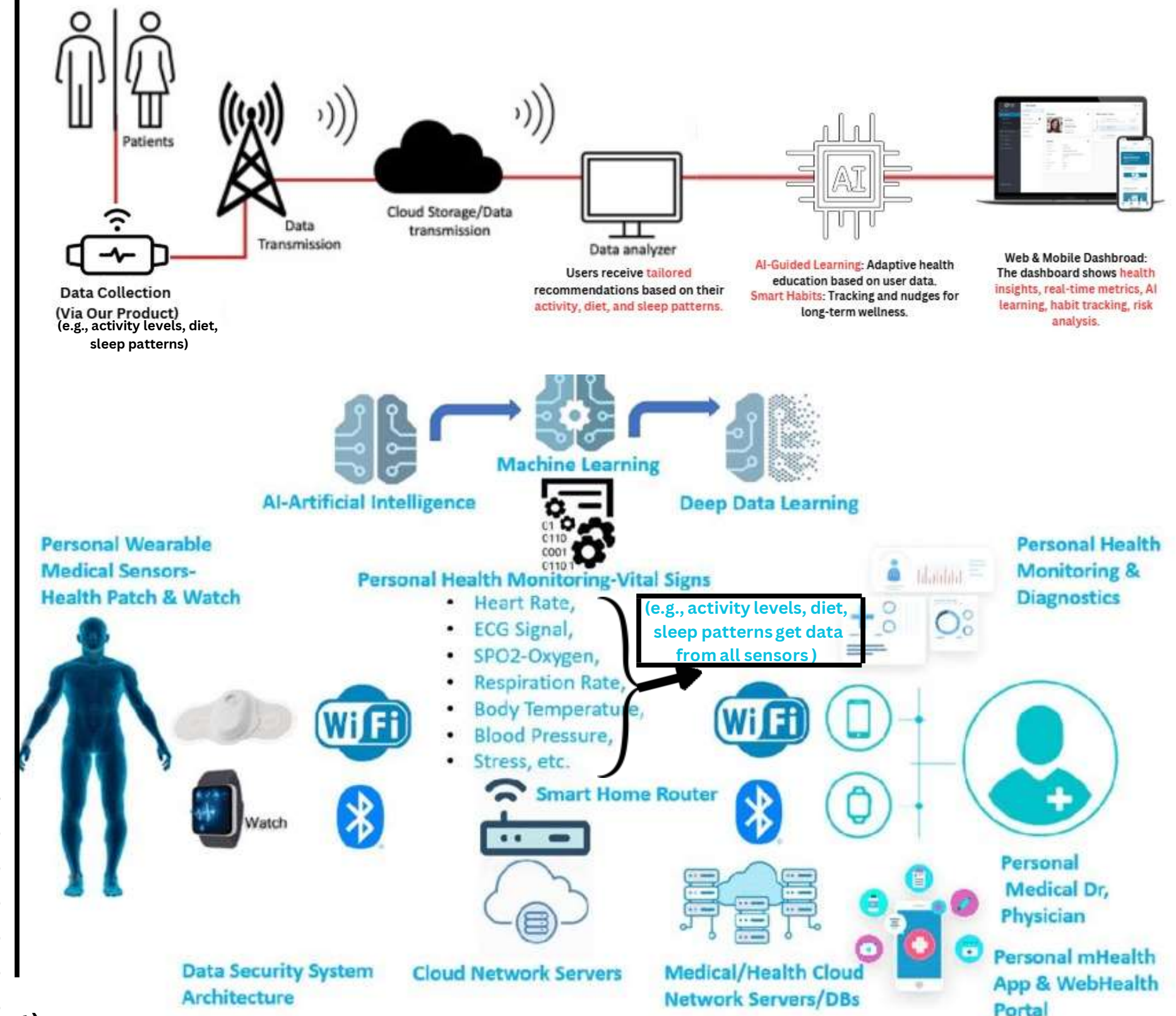
End-to-End Flow:

Sensors → IoT Gateway → Cloud → AI Processing → Insights to Users

Hardware components used:

Sensor Name	Type	Sensor Name	Type	Sensor Name	Type
Raspberry Pi 4	Microcontroller	MAX30205	Temp. Sensor	ADXL345	Actigraphy Sensor
Max30102	PPG Sensor	ADS1292R	ECG Sensor	Pneumotrace	Respiration Sensor
MPU6050	Gyroscope	GSR Sensor	GSR Sensor	Libre Freestyle	CGM Sensor
SGP30	Air Quality Sen.	NeuroSky TGAM	EEG Sensor	DFRobot Gravity TDS	Hydration Sensor
ML8511	UV Sensor	TSL2561	Blue Light Sen.	LoRa Module (SX1276)	Comm. Module

Fetal Monitoring : Doppler Ultrasonic Sensor(Used to detect the heart beat and movement)



How Unique is it?

SMARTWATCHES



- Basic tracking (heart rate, steps, sleep).
- No direct emergency response features.
- Limited communication with healthcare providers.
- Local storage and basic tracking.
- Limited feedback and insights
- Basic health tracking and fitness monitoring.
- Expensive but mainly focused on fitness and daily health tracking
- May have limited accuracy, especially in sensors for blood pressure or ECG (due to form factor)

Example:

Apple Watch Series 8 - Price: ₹49,900 - ₹75,900

Samsung Galaxy Watch 6 - Price: ₹29,990 - ₹49,990

Garmin Fenix 7X Solar - Price: ₹89,990 - ₹1,15,990

Fossil Gen 6 - Price: ₹22,995 - ₹27,995

OUR PROJECT



- Comprehensive monitoring (ECG, blood pressure, oxygen levels, etc.)
- Real-time emergency alerts to caregivers, healthcare providers, and close contacts
- Personalized care plans for medicine, physiotherapy, and nutrition.
- Seamless integration with doctors, pharmacies, blood banks, and ambulance services
- Cloud-based storage with advanced real-time analysis and early risk detection.
- Continuous health feedback, including risk alerts and proactive care.
- High accuracy with medical-grade sensors for precise health data (blood pressure, ECG, etc.)
- Affordable, providing comprehensive healthcare monitoring for patients with specialized needs.
- Focus on preventing health complications and improving long-term patient outcomes.

FEATURES



1) ADDITION OF CGM (CONTINUOUS GLUCOSE MONITOR)

CGM systems use electrochemical glucose sensors to measure glucose levels in the human body. These sensors send a signal through the transmitter to the person through an app.

ADVANTAGES:

- Helps Avoid Diabetic Ketoacidosis (DKA)
- Prevents Hypoglycemia & Hyperglycemia
- Better Insulin Dosing/ Insulin pump integration
- Real-time Monitoring
- Stable glucose levels lower the risk of heart disease, kidney damage, nerve damage

2)Integration of the Auto Injectors

Pre-filled medications is inserted using a spring loaded mechanism with one button operation through the band for easier navigation and insertion

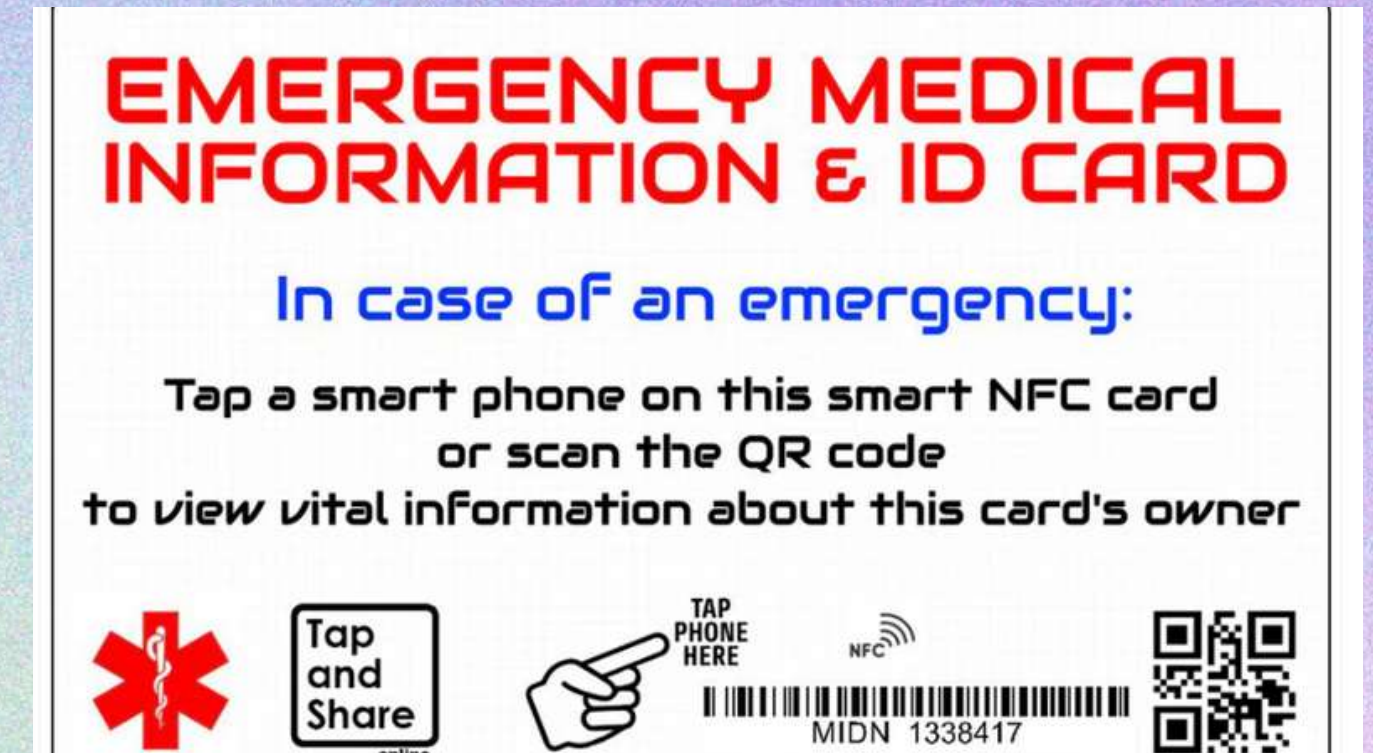
ADVANTAGES:

- **EMERGENCY AUTO INJECTIONS:** Ephiphrine for allergic reactions, diazapham for seizure attacks, naxoline for opiium overdoses
- **SMART-AUTO INJECTORS:** Bluetooth-connected/NFC devices track injection history and send reminders, AI-driven auto-injectors that adjust dosage based on real-time patient data
- **Portable & Ready-to-Use:** Convenient for emergency situations.



ADDITION OF NFC:

- NFC programmed to show the medical history of the person when admitted in emergency so that doctors can give treatment as per the patients need
- Use of the the nfc in showing the data when a person is in need of emergency to alert their family members



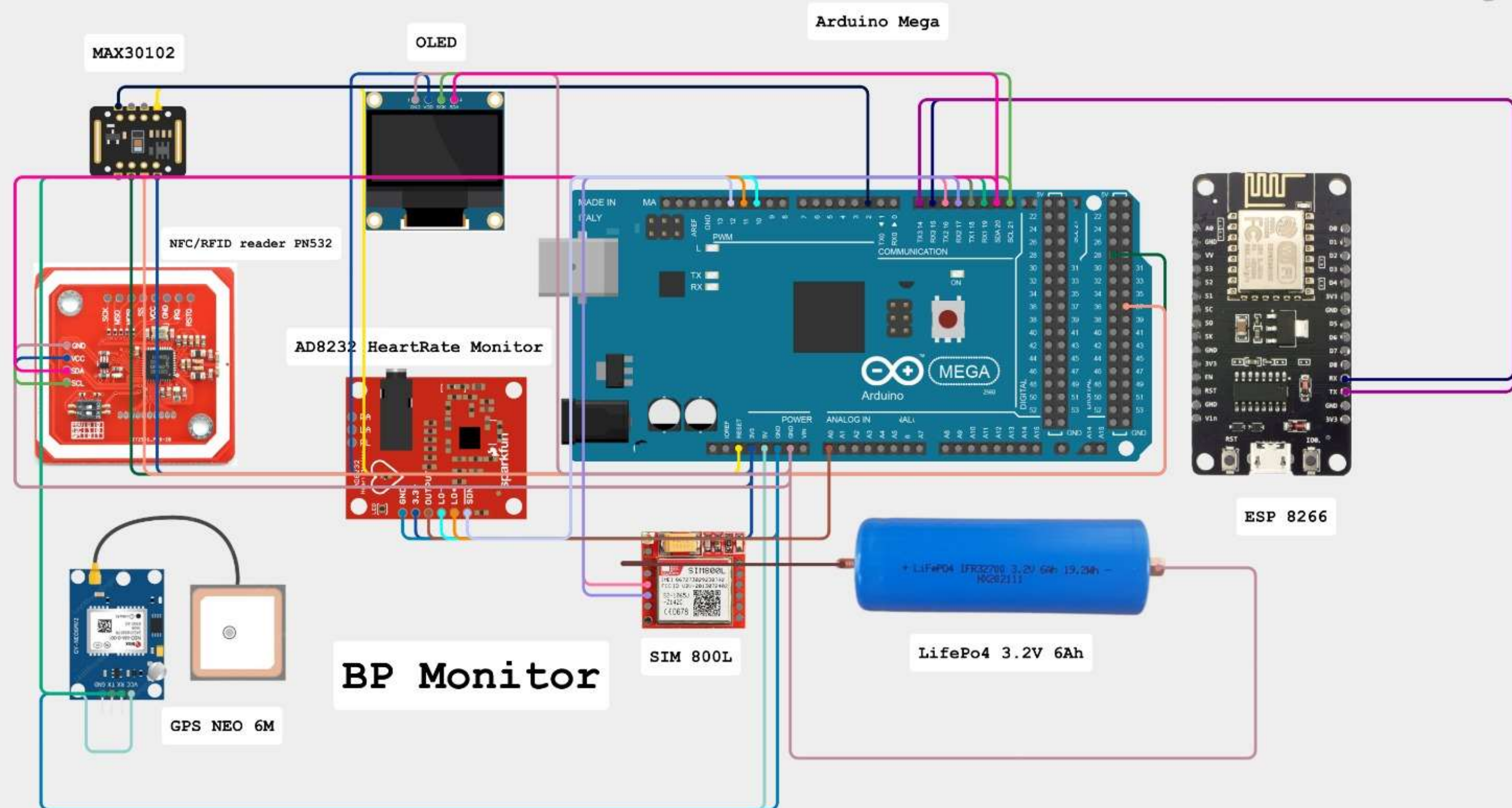
capacitive touch sensor



**It is used detect the whether the band is placed in the arm or not .
If not the the band will go to the Low energy Consumption mode**

Schematic Diagram

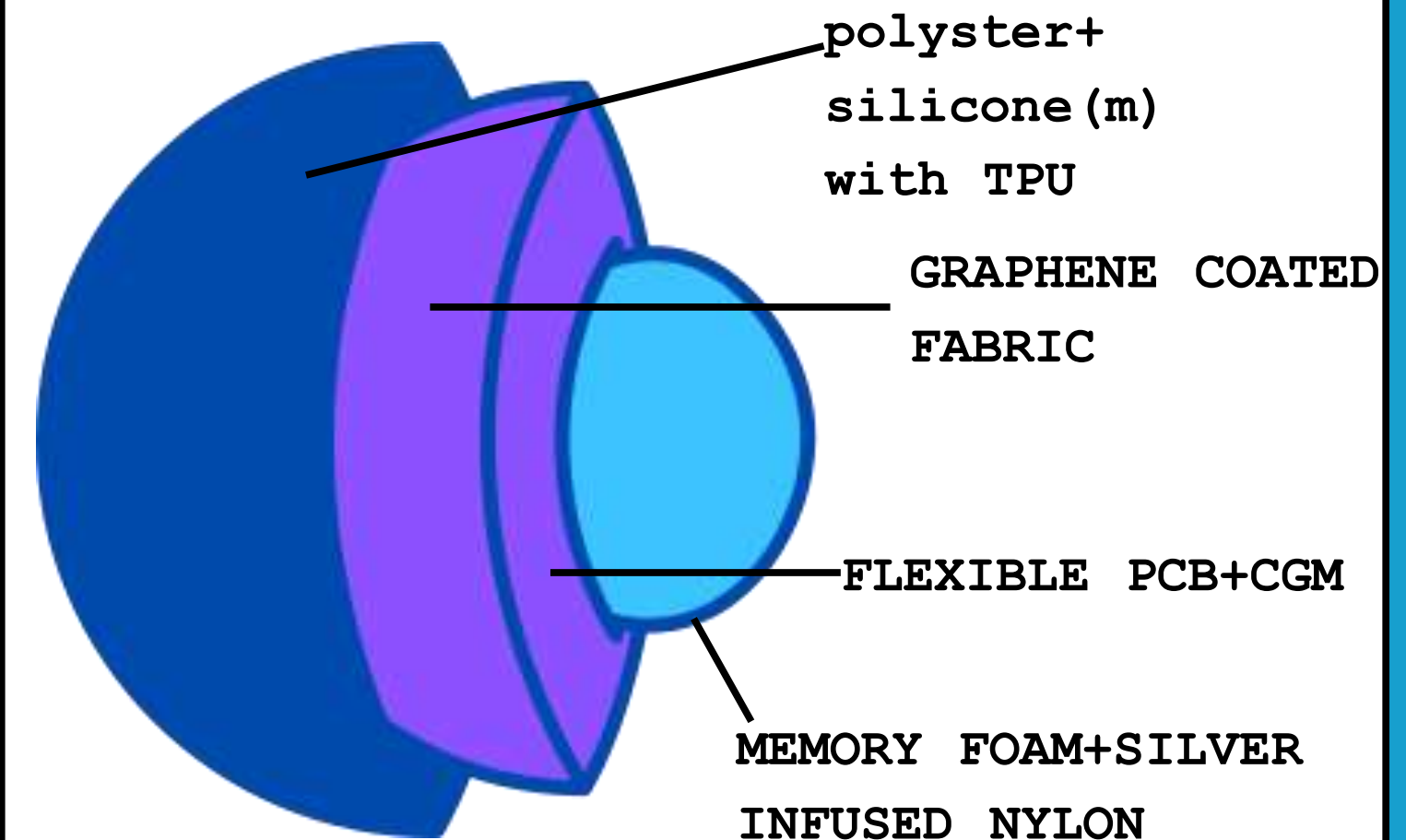
Circuit Designer



MATERIAL LAYOUT

THE DIFFERENT MATERIALS USED ARE GIVEN WITH ITS CORRESPONDING USES:

- Outer Layer-polyster+medical grade silicon with TPU (THERMOPLASTIC POLYURETHANE) for wear and tear,waterproffing,dust and moisture
- Middle Layer-graphene coated fabric for better heat dissipation $\sim 5300 \text{ W/m}\cdot\text{K}$
- Flexible PCB+CGM
- INNER CORE-memory foam with a silver infused nylon for anti bacterial properties and sweat absorption



Battery Optimization Strategies

Component	Voltage	Est. Current Draw (Typical)	Notes
MAX30102	1.8–3.3V	~600 μ A (avg), 1.1 mA (peak)	Intermittent sampling helps
DIY BP (6V 2000mAh)	6V	High draw (motor pump based)	Optimize usage only when needed
GPS (Steps Tracker)	3.3V	~25-30 mA	GPS is power-hungry, avoid continuous mode
AD8232 (Stress/ECG)	3.3V	~170 μ A	Very low-power
DS18B20 (Temp)	3–5.5V	~1.5 mA (during conversion)	Use sleep mode between reads
SIM900A (GSM)	3.4–4.4V	~2A peak during TX, ~20 mA idle	Very power-intensive

❖ Duty Cycling : Turn sensors on only when needed. For example:

- ✓ Temp & HR → every 10 seconds
- ✓ GPS → every 5–15 minutes (unless in emergency)
- ✓ GSM → transmit in bulk, not real-time unless alerting

❖ Low Power Modes (MCU + Sensors):

- ✓ Use ESP32’s deep sleep mode.
- ✓ MAX30102 and AD8232 support low-power or standby modes.

❖ Capacitive Arm Detection Sensor (as in your design) : If the band is removed, put entire system to deep sleep to conserve battery.

❖ Use Boost/Buck Converters Efficiently :
Avoid linear regulators (e.g., LM7805). Use switching regulators like TPS61200 for 3.3V from Li-ion safely and efficiently.

🔋 Battery Alternatives (Explosion-Safe & Wearable-Safe)

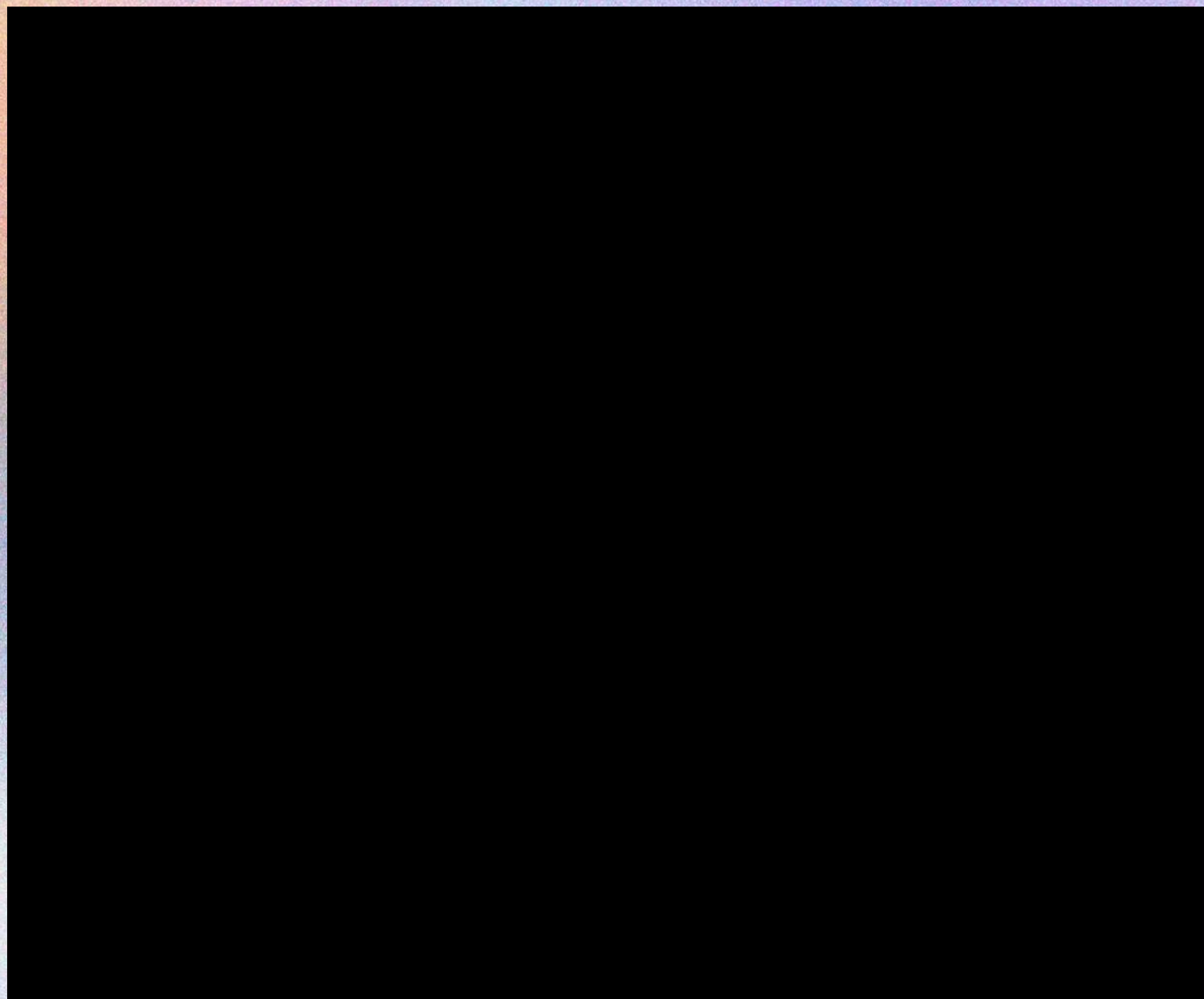
Battery Type	Pros	Cons	Safe Option
Li-Po (Lithium Polymer)	Flat, flexible, lightweight	Risk of swelling/fire	✅ <i>Use with BMS, temp cut-off & enclosure</i>
LiFePO4 (Lithium Iron Phosphate)	Very safe, thermally stable	Slightly heavier, lower energy density	✅ <i>Best for safety-critical apps</i>
Zinc-Air (Rechargeable)	Ultra-safe, no explosion risk	Shorter lifespan	✅ <i>If very lightweight & low draw</i>
Coin Cells (CR2032, etc.)	Cheap, easy for ultra-low-power	Can't support GSM or GPS	❌ <i>Not suitable here</i>

🔒 Tips for Safety in Battery Design

- ❖ Use a fire-retardant enclosure (e.g., TPU + silicon).
- ❖ Include a thermal fuse or NTC thermistor to shut off charging when the battery overheats.
- ❖ Add reverse polarity and over-discharge protection using TP4056 + DW01 combo.
- ❖ Avoid charging during usage—charge during rest periods only.
- ❖ Use non-conductive, shock-absorbing padding between battery and PCB layers.

OURS VS CONSUMERS HEALTH BAND

Sensor	Your Smart Band	Consumer Health Bands	Accuracy	Best Use Case
ECG (Heart Monitoring)	MAX30003, AD8232	Apple Watch, Fitbit Sense	✅ High	Detecting heart conditions (arrhythmia, HRV)
Heart Rate (HR)	ECG-based HR (Accurate)	PPG-based HR (Light-based)	⚠️ Moderate	Fitness tracking, basic HR monitoring
SpO2 (Blood Oxygen)	MAX30102	Apple Watch, Garmin, Fitbit	⚠️ Moderate	Sleep tracking, altitude sickness
Blood Pressure (BP)	AI-based BP Estimation	Samsung Galaxy Watch (PPG-based)	⚠️ Low	Trend monitoring, not medical use
Temperature	Infrared + Thermistor	Fitbit Sense, Huawei, WHOOP	⚠️ Moderate	Fever tracking, stress monitoring
Fall Detection	AI + Accelerometer	Apple Watch, Garmin	⚠️ Moderate	Elderly monitoring, emergency alerts
Sleep Tracking	HRV + AI-based analysis	Fitbit, WHOOP, Oura Ring	⚠️ Moderate	Sleep stages, wellness tracking



TI-BAND

FeaturesDashboardRecomendationTaskAboutContact

🍽️ Weekly Food Plan (Healthy Eating)

This meal plan supports energy levels, gut health, and better sleep.

🍳 Breakfast (6:30 AM)

Monday: Oats + Almonds + Greek Yogurt

Tuesday: Scrambled Eggs + Avocado Toast

Wednesday: Smoothie (Banana + Berries + Protein)

Thursday: Poha / Idli with Coconut Chutney

Friday: Whole Wheat Pancakes + Peanut Butter

Saturday: Sprouts & Chia

🍲 Lunch (12:00 PM)

Monday: Brown Rice + Grilled Chicken + Salad

Tuesday: Quinoa + Paneer Curry + Raita

Wednesday: Chapati + Dal + Vegetables

Thursday: Millet Bowl + Boiled Eggs + Greens

Friday: Whole Wheat Pasta + Veggies

Saturday: Dal Khichdi + Curd

🍵 Evening Snack (6:00 PM)

Monday: Nuts & Green Tea

Tuesday: Fruit + Yogurt

Wednesday: Roasted Chickpeas

Thursday: Dark Chocolate + Herbal Tea

Friday: Sweet Potato Chaat

Saturday: Mixed Seeds & Dry Fruits

Sunday: Soup + Whole Wheat Crackers

🌙 Dinner (8:00 PM)

Monday: Lentil Soup + Grilled Vegetables

Tuesday: Paneer Bhurji + Roti

Wednesday: Grilled Fish + Steamed Veggies

Thursday: Chickpea Salad + Yogurt

Friday: Moong Dal Khichdi + Buttermilk

Saturday: Light Dosa + Sambar

Sunday: Vegetable Stir-Fry +

TI-BAND

FeaturesDashboardRecomendationTaskAboutContact

🌡️ Body Temperature

36.9°C

Normal

Past 24hr: +0.2

❤️ Heart Rate

73 bpm

Normal

Past 24hr: Stable

🩸 Blood Pressure

117/76

Normal

Past 24hr: -1.0

View History

👤 Oxygen Saturation

98%

Normal

Past 24hr: +1.0

👤 Steps

Default

keep walking

Past 24hr: Avg=0

📈 Blood Glucose

Default

Normal

Past 24hr: -5 mg/dl

🔥 CALORIES

Default

Normal

Past 30d: 0

📊 ECG

Normal

Sinus Rhythm

Last check: 2h ago

TI-BAND

FeaturesDashboardRecomendationTaskAboutContact

📅 Daily Schedule

Time	Activity	Notes
3:00 AM	Wake up & Hydrate	Drink 500 ml of warm water to activate digestion.
3:30 AM - 4:00 AM	Meditation & Deep Breathing	Helps wake up your brain & improve focus.
4:00 AM - 4:30 AM	Light Workout (Stretching/Yoga)	Keep the body active without over-exertion.
4:30 AM - 5:00 AM	Morning Tea / Herbal Drink	Green Tea / Lemon Honey Water / Turmeric Tea
5:00 AM - 6:00 AM	Work / Study / Productivity	Best time for deep focus.
6:00 AM - 7:00 AM	Healthy Breakfast	Protein + Good Fats + Fiber (See meal plan below)
7:30 AM - 12:00 PM	Work / Meetings / Classes	Take small breaks every 2 hours.
12:00 PM - 12:30 PM	Lunch	Balanced meal with carbs, proteins, veggies.

Enter Heart Rate: 80

Enter SpO2: 100

Enter Steps: 1000

Enter Sleep (hrs): 7

Enter Stress Level (Low, Moderate, High): Moderate

Enter Calories Burnt: 7.3

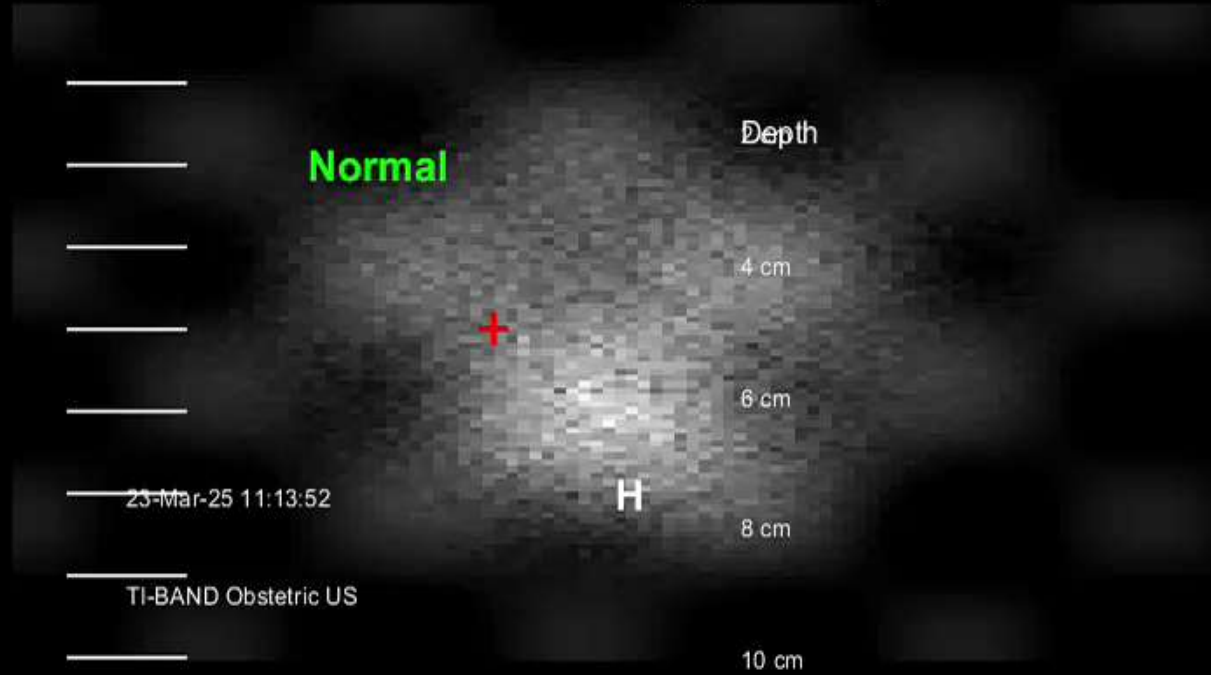
Enter Systolic BP: 123

Enter Diastolic BP: 72

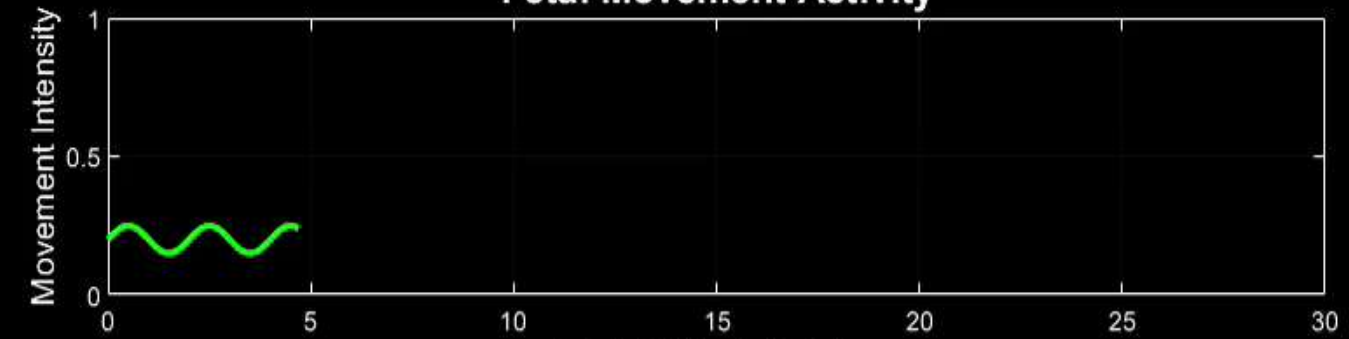
Predicted Health Condition: ['Sedentary']



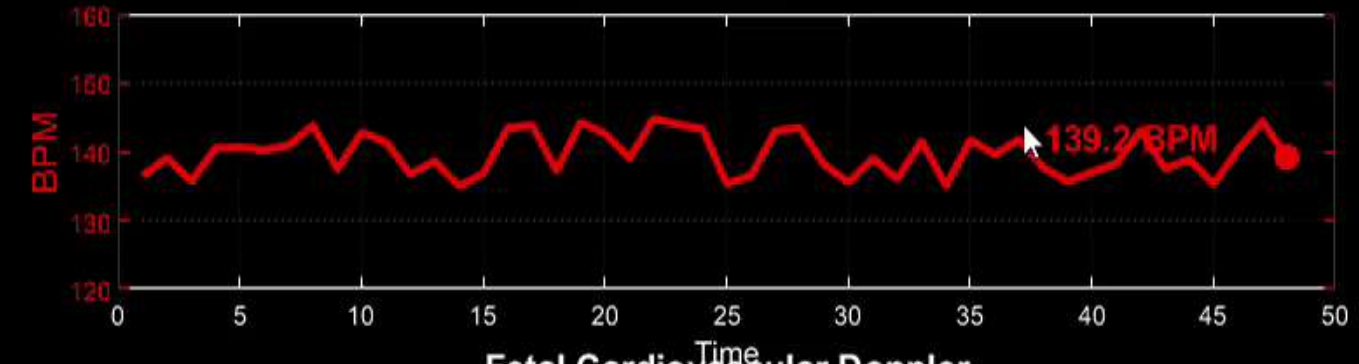
Fetal Ultrasound (28 weeks)



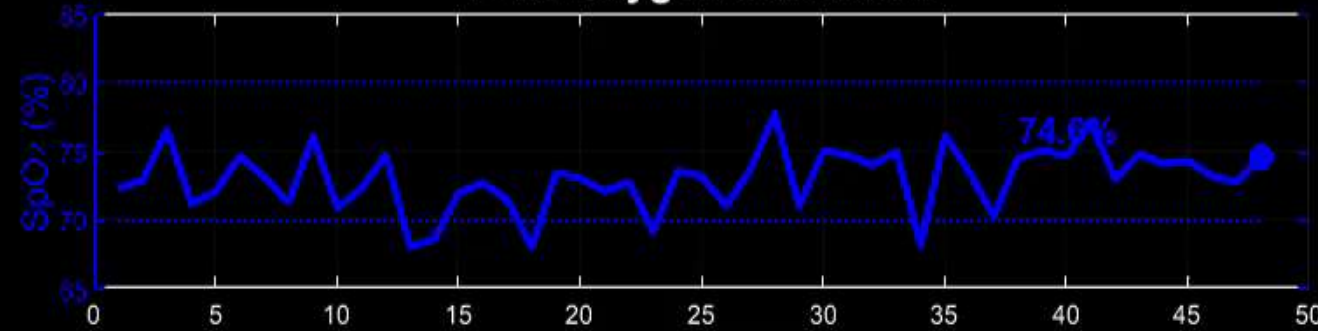
Fetal Movement Activity



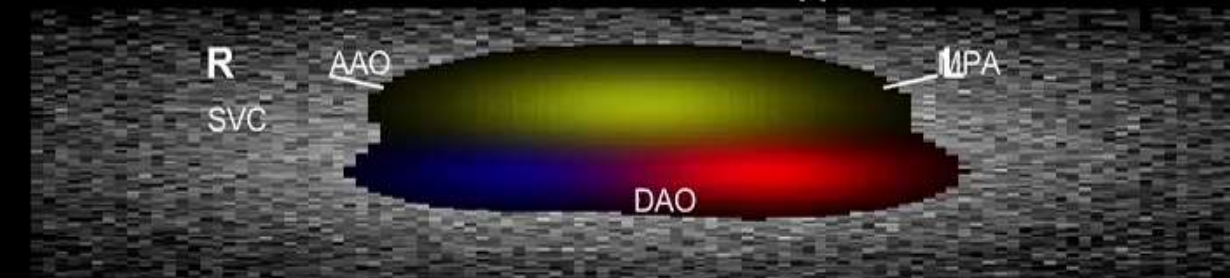
Fetal Heart Rate



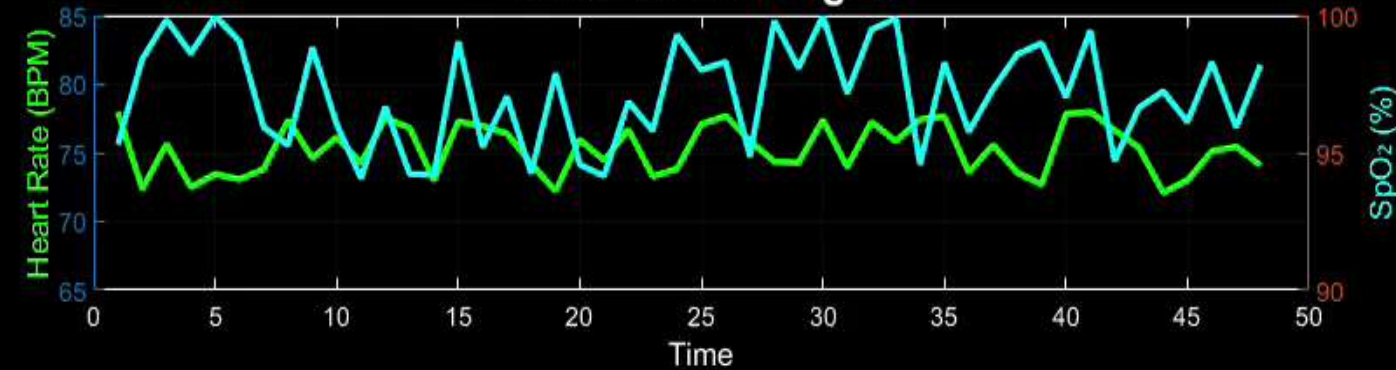
Fetal Oxygen Saturation



Fetal Cardiovascular Doppler



Maternal Vital Signs



Fetal Assessment Summary

CURRENT STATUS:

Movement: normal
Head Position: upward
Kicks Detected: 0
Activity Level: LOW

FETAL VITALS:

HR: 139.2 BPM
SpO2: 74.6%

FETAL ASSESSMENT:

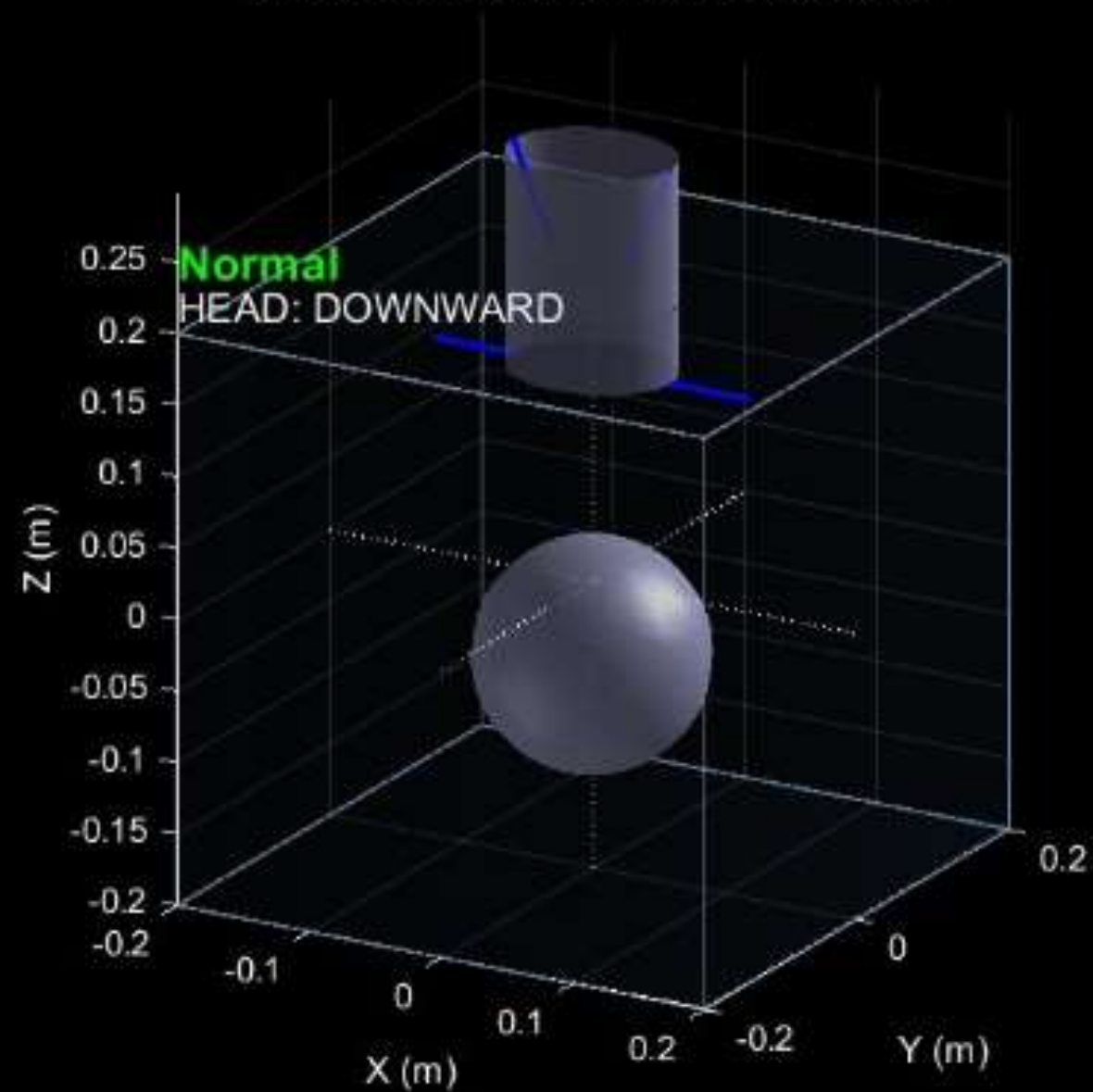
Gestational Age: 28 weeks
Biophysical Profile: 4/10
Est. Fetal Weight: 2300 g
Presentation: Breech

SESSION INFO:

Session time: 4.7 s
Data points: 48

23-Mar-25 11:13:52

3D Fetal Movement Simulation



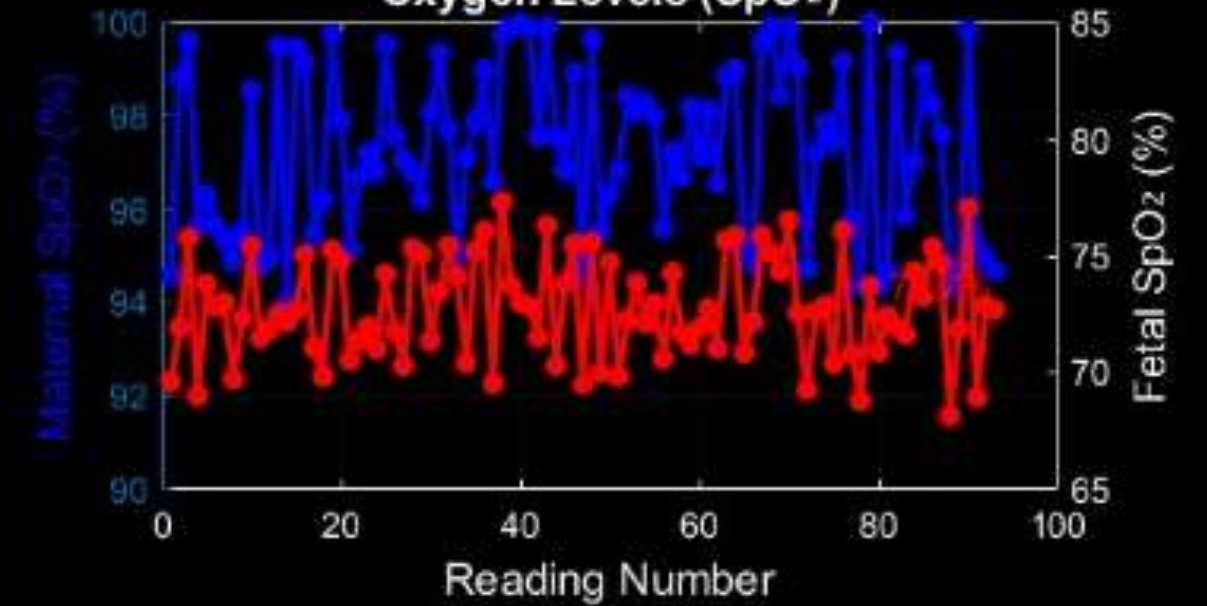
CURRENT STATUS:

Movement: normal
Head Position: downward
Kicks Detected: 2
Activity Level: MEDIUM

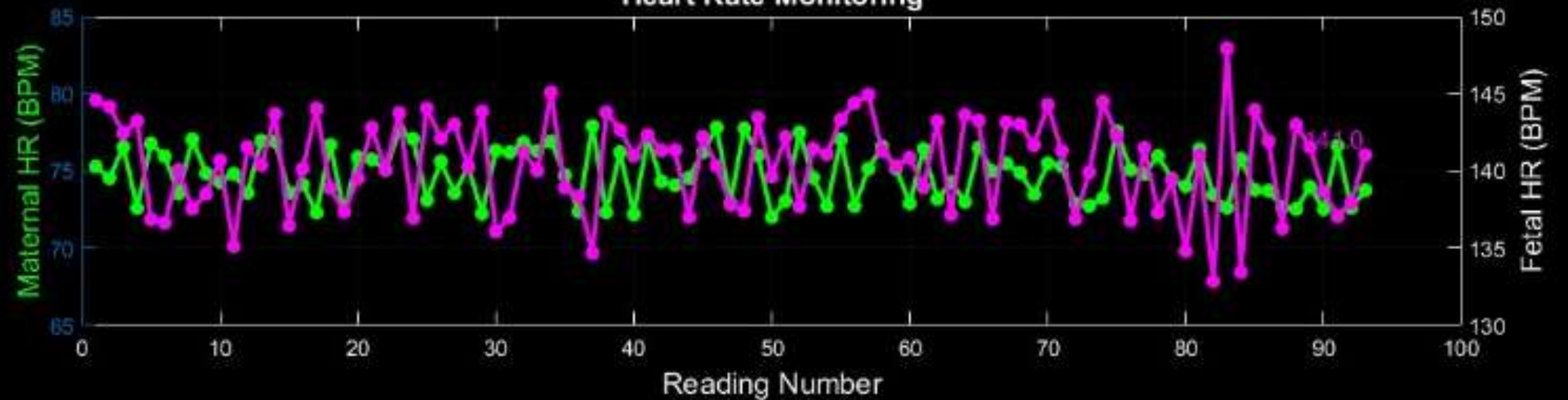
Fetal Movement Activity



Oxygen Levels (SpO₂)



Heart Rate Monitoring



Comprehensive Fetal Monitoring System

VITAL SIGNS:

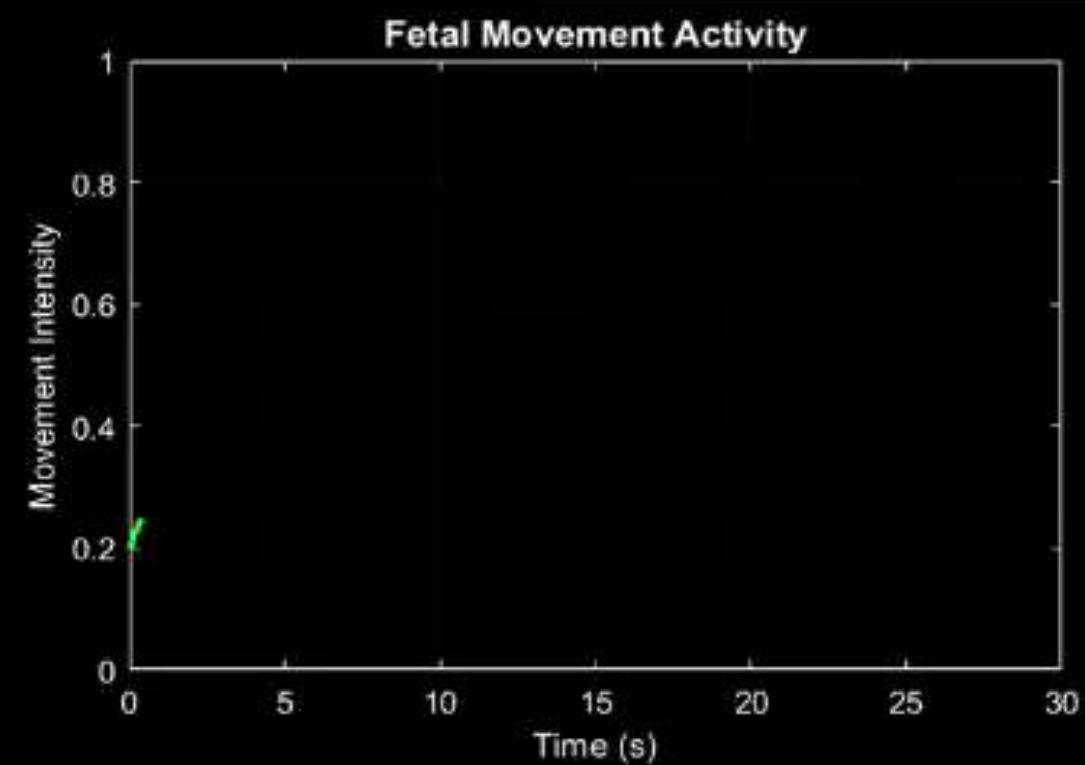
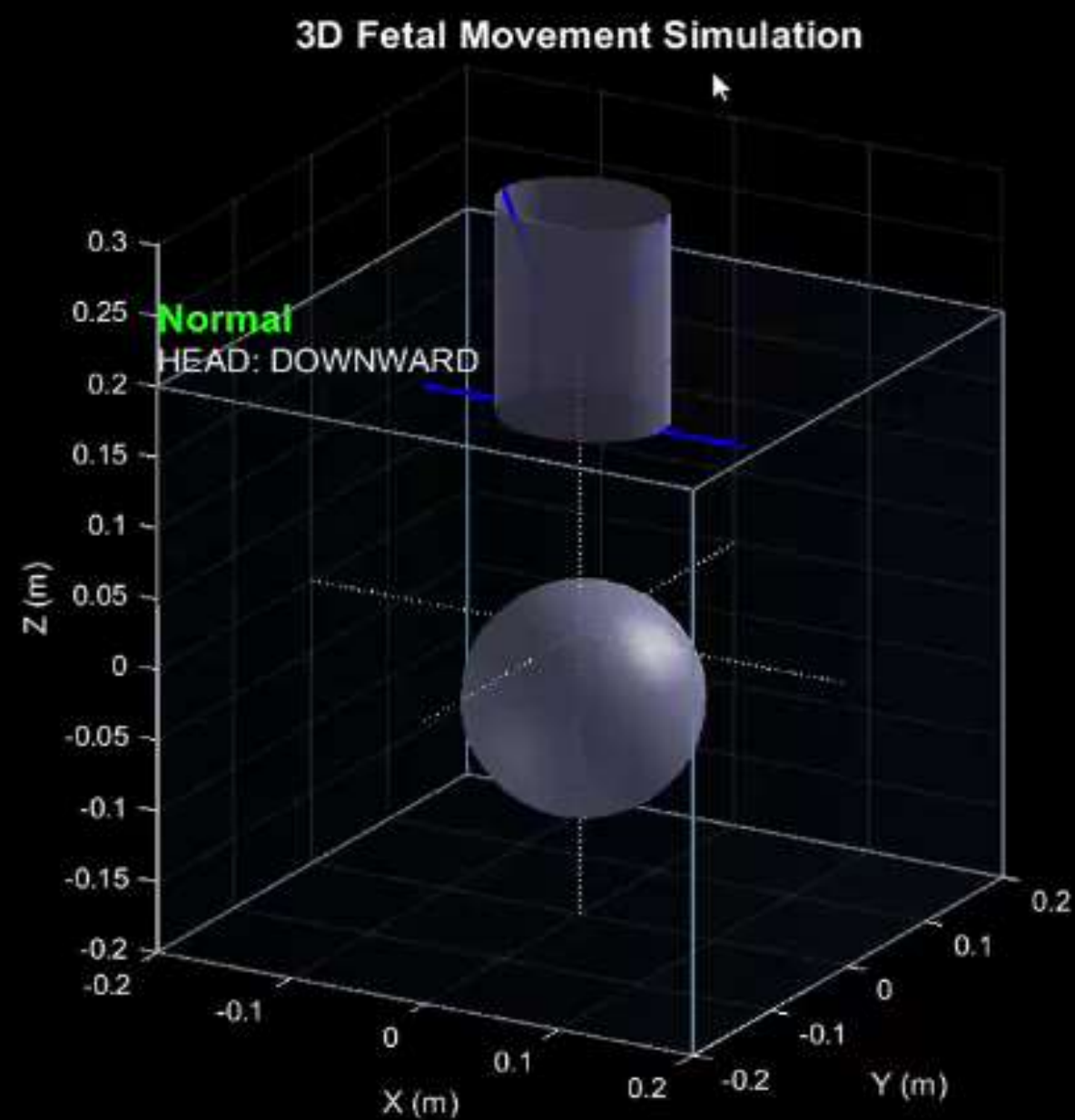
Oxygen Levels (SpO₂):
Maternal: 94.7%
Fetal: 72.7%
Heart Rate:
Maternal: 73.7 BPM
Fetal: 141.0 BPM

POSITION DATA:

X: -0.000 m
Y: -0.001 m
Z: 0.100 m

SESSION INFO:

Session time: 9.2 s
Data points: 93



Fetal Movement Monitor

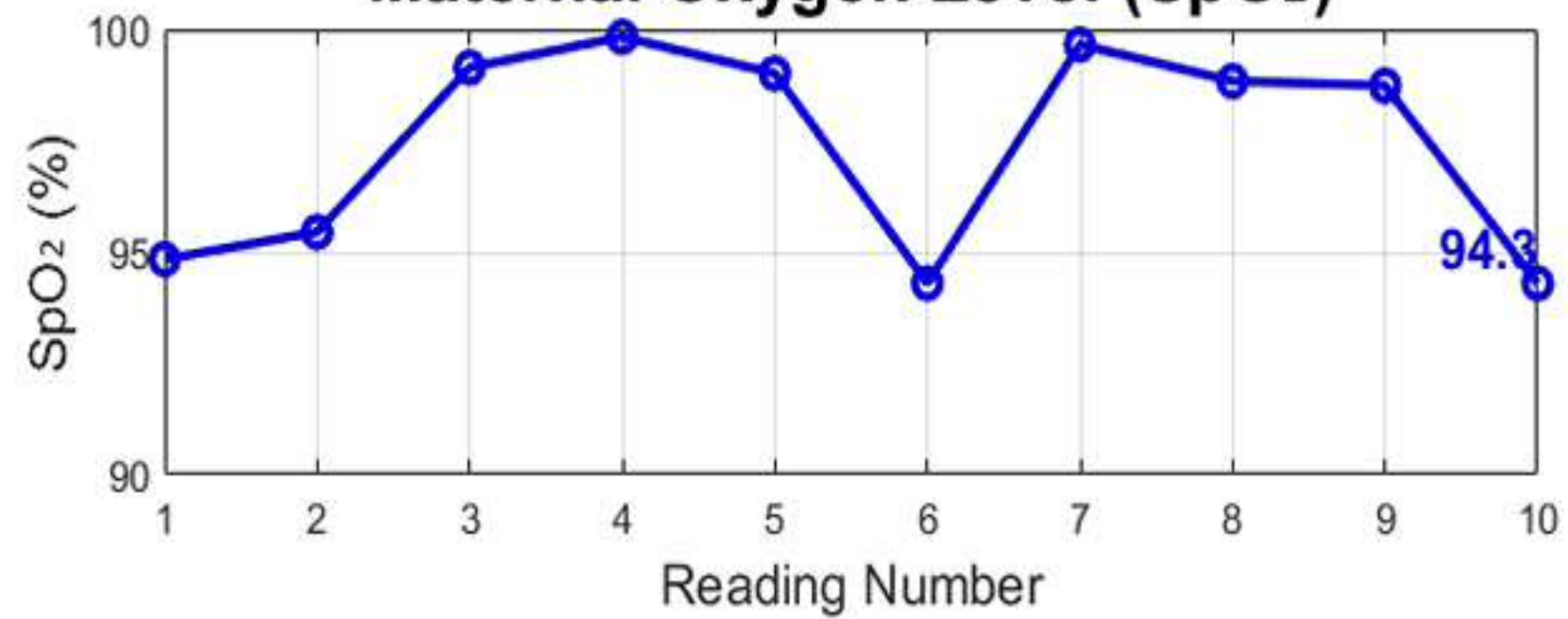
CURRENT STATUS: MOVEMENT STATS:
Movement: normal **Kicks: 0**
Head Position: downward **Activity: LOW**
POSITION DATA: Session time: 0.4 s
X: 0.001 m Data points: 5
Y: 0.003 m
Z: 0.102 m

22-Mar-25 22:39:11

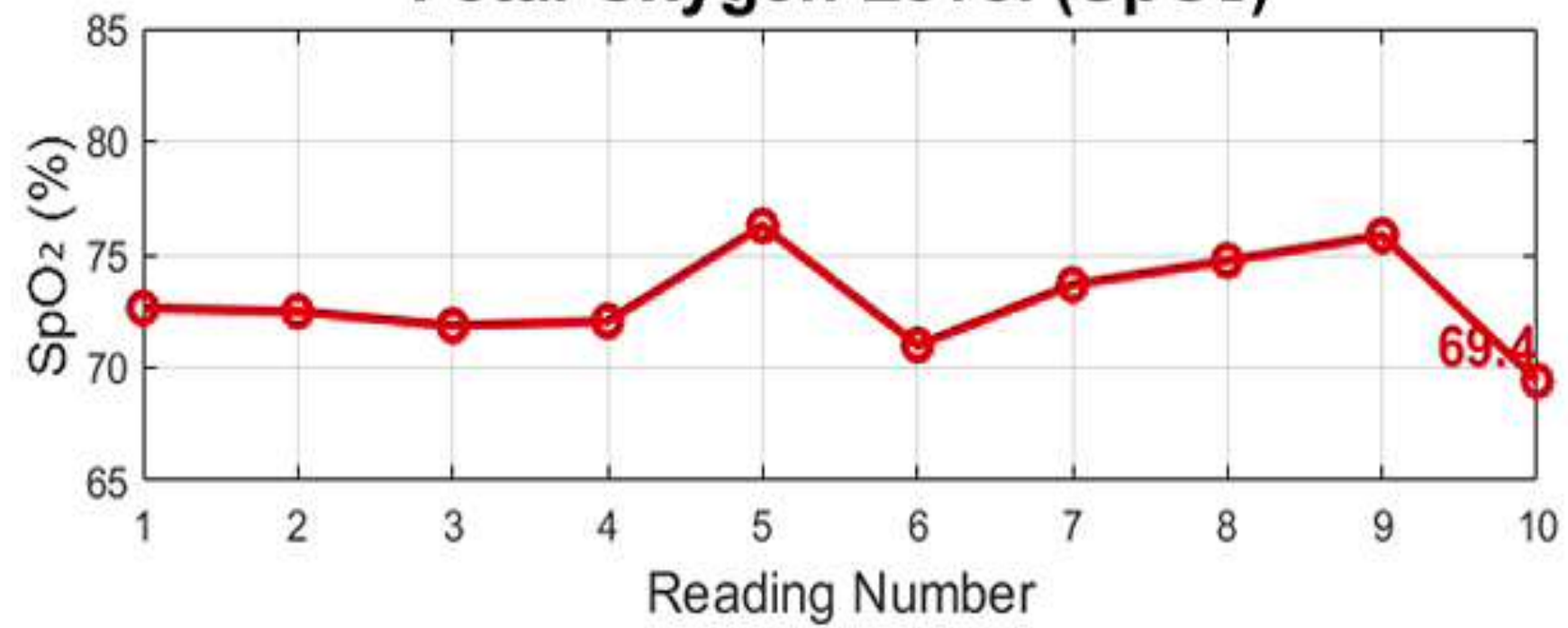
Data From the TI BAND



Maternal Oxygen Level (SpO₂)



Fetal Oxygen Level (SpO₂)



COMPETITORS



Dexcom has launched a proprietary Generative AI (GenAI) platform integrated with its glucose biosensing technology.

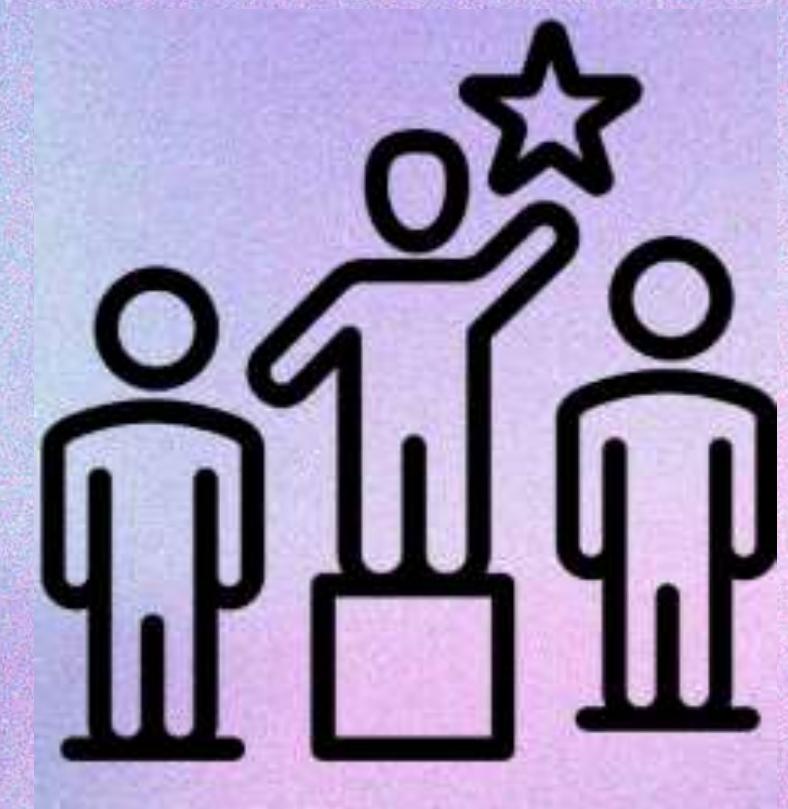
Lark Health provides an AI-powered platform offering a coaching for health issues



Glooko is a comprehensive diabetes management platform that integrates with over 95% of diabetes devices

COMPETITIVE ADVANTAGE

- **IMPLEMENTATION OF COMPREHENSIVE HEALTH EDUCATION**
- **BEHAVIORAL SCIENCE INTEGRATION WHERE IT ADAPTS AND CHANGES ITSELF ACCORDING TO THE PERSON**
- **PRIVACY-FOCUSED DATA MANAGEMENT**
- **NFC INTEGRATION TO PROVIDE A BETTER AND EASIER TRACK OF MEDICAL RECORDS AND ASSIST DURING MEDICAL EMERGENCIES**
- **USE OF AUTO INJECTOR DURING MEDICAL EMERGENCIES OR FOR IMMEDIATE ACTION HELPS SAVING LIFE**



SOCIAL IMPACT AND FEASIBILITY

Social Impact



Reduces Maternal and Infant Mortality

- Enables early detection of complications (e.g., abnormal fetal heart rate; high BP)
- Sends instant alerts to doctors and caregivers for timely intervention



Promotes Preventive Healthcare

- Shifts focus from reactive to proactive care by identifying risk factors early
- AI-based insights help in preventing severe health complications



Empowers Women with Health Awareness

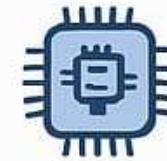
- Provides personalized health recommendations based on real-time monitoring
- Educates mothers about nutritional needs, fetal development, and warning signs



Bridges the Healthcare Gap in Rural Areas

- IoT-enabled wearable devices allow remote health monitoring for women with limited access to hospitals
- Reduces dependency on physical checkups, ensuring continuous monitoring from home

Feasibility



Technical Feasibility

- Uses AI, cloud computing and IoT-enabled wearable sensors for seamless data processing
- Integrated real-time monitoring, risk prediction, and automated alerts emergencies



Data Security & Privacy

- End-to-end encryption (AES-256) ensures data protection
- Compliance with GDPR & HIPAA to prevent unauthorized access



Scalability & Industry Readiness

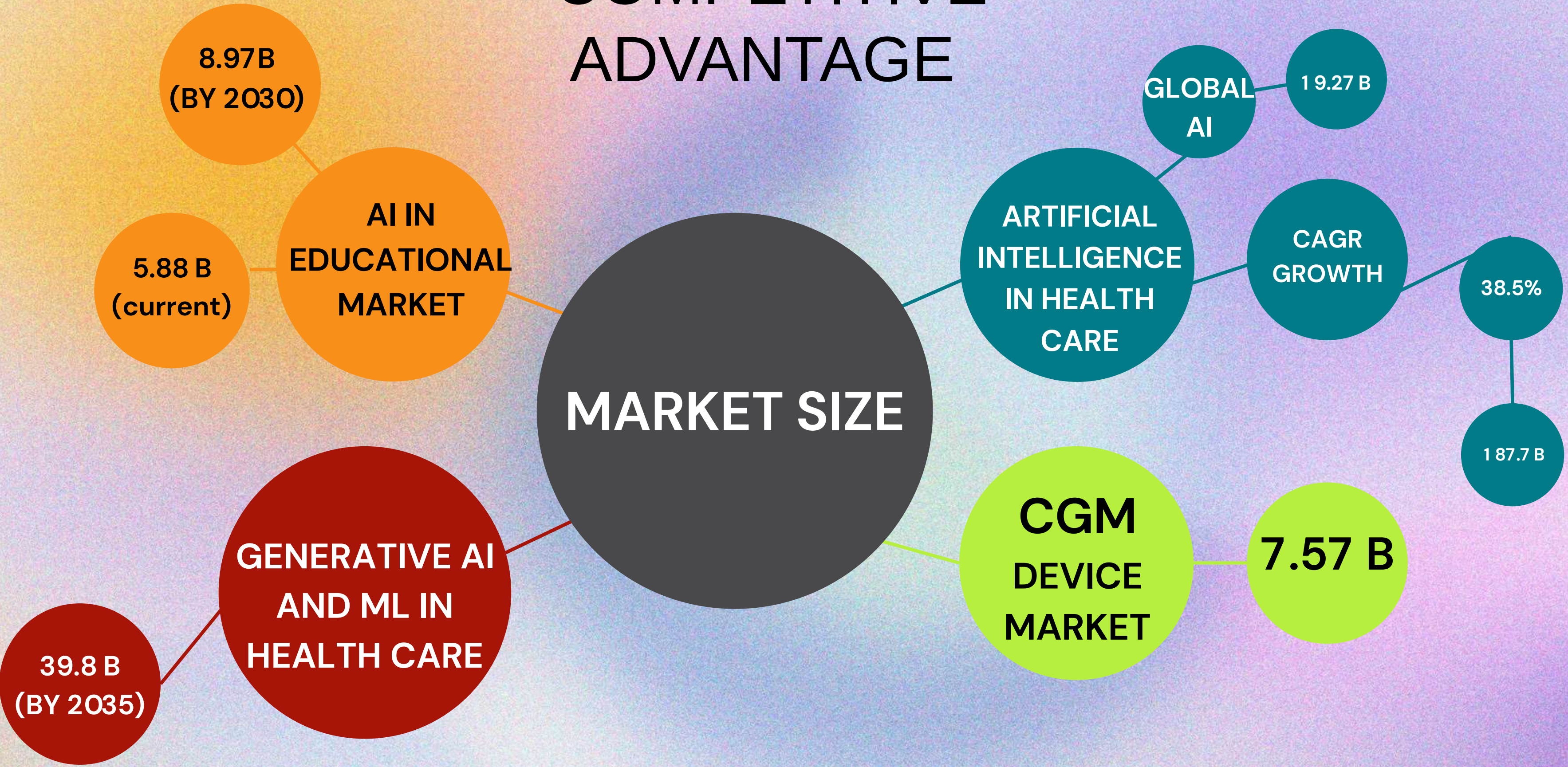
- Modular architecture supports future expansion & upgrades
- Can be marketed to pregnant women; women, clinics, and rural healthcare centers



Cost Efficiency

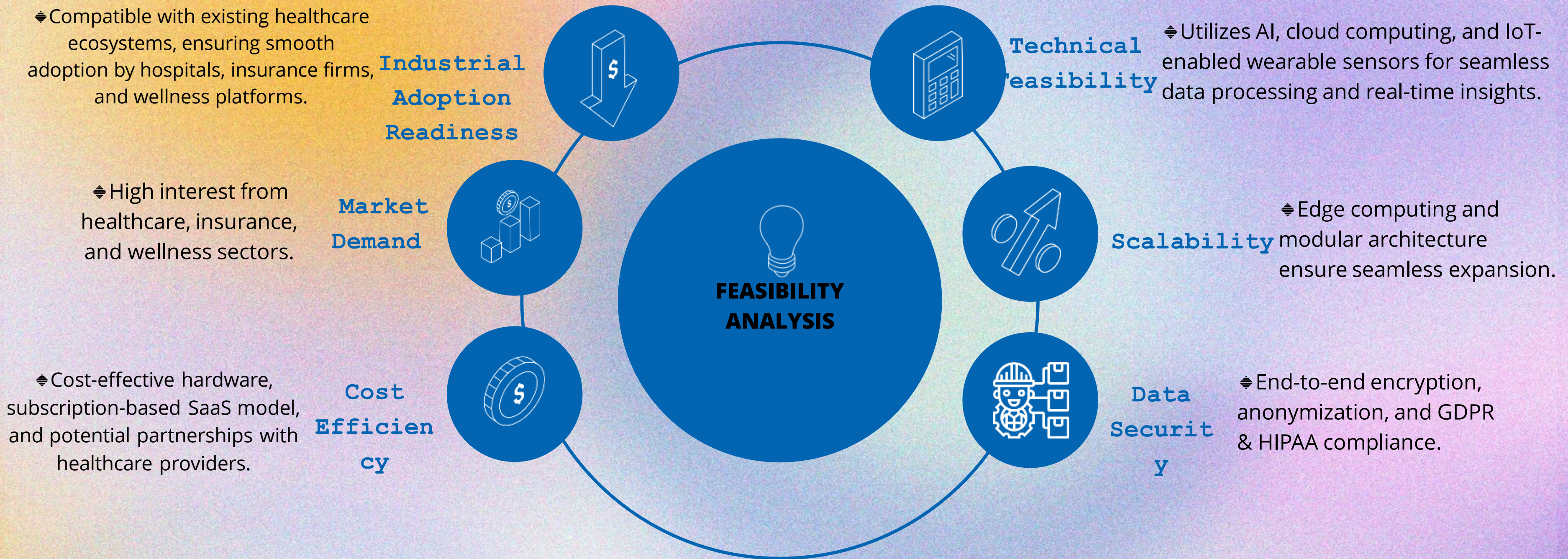
- Affordable alternative to hospital-grade monitors
- Reduces hospital visits and emergency admissions, saving medical costs
- Possible partnerships with NGOs & government health programs for large-scale distribution

COMPETITIVE ADVANTAGE



SOURCES:GRANDVIEWRESEARCH.COM
ROOTSANALYSIS.CO

Feasibility Analysis



The **HealthBand**, priced at **₹18,000**, stands out as one of the **most portable and advanced health monitoring devices** in the market today. With **AI-driven analytics, real-time cloud connectivity, and high-precision sensors**, it offers a **lightweight** yet powerful solution for **personalized health tracking**. Its compact design, cross-platform accessibility, and future **scalability** make it **a cost-effective and cutting-edge choice for users seeking real-time health insights**.

TARGET MARKET

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graph TD; TM[TARGET MARKET] --> CM[CORE MARKET]; TM --> GS[GROWING SEGMENT]; TM --> MME[MASS MARKET EXPANSION]; CM --> CM_L["• PATIENTS WHO REQUIRE MEDICAL ATTENTION 24/7<br>• PEOPLE AT RISK OF OF SUDDEN HEALTH EMERGENCY<br>• INDIVIDUALS SUCH AS ELDERLY PEOPLE, OWEWEIGHT, KIDS AND PREGNANT WOMEN"]; GS --> GS_L["• HEALTH-CONSCIOUS CONSUMERS<br>• FITNESS ENTHUSIASTS, ATHLETES, AND LONGEVITY-FOCUSED INDIVIDUALS<br>• USERS OF WEARABLE HEALTH TECH LIKE APPLE WATCH, FITBIT, WHOOP, OURA RING"]; MME --> MME_L["• PEOPLE LOOKING FOR NUTRITION GUIDANCE<br>• INDIVIDUALS STRUGGLING WITH INSULIN OR OTHER REGULAR MEDICATION"];
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CORE MARKET

- PATIENTS WHO REQUIRE MEDICAL ATTENTION 24/7
- PEOPLE AT RISK OF OF SUDDEN HEALTH EMERGENCY
- INDIVIDUALS SUCH AS ELDERLY PEOPLE, OWEWEIGHT, KIDS AND PREGNANT WOMEN

GROWING SEGMENT

- HEALTH-CONSCIOUS CONSUMERS
- FITNESS ENTHUSIASTS, ATHLETES, AND LONGEVITY-FOCUSED INDIVIDUALS
- USERS OF WEARABLE HEALTH TECH LIKE APPLE WATCH, FITBIT, WHOOP, OURA RING

MASS MARKET EXPANSION

- PEOPLE LOOKING FOR NUTRITION GUIDANCE
- INDIVIDUALS STRUGGLING WITH INSULIN OR OTHER REGULAR MEDICATION

Business Model Canvas

Category	Description
Customer Segments	Pregnant women (urban + rural), Hospitals & Clinics, Pharmacies, Government health agencies
Value Proposition	Continuous maternal + fetal monitoring, Emergency alerts, AI insights, NFC access, Auto-injector, gamification, cloud sync
Customer Relationships	App dashboard, Subscriptions, Chatbot assistance, AI recommendations
Revenue Streams	Device sale (Rs.18,000), Hospital SaaS, NFC commissions, Govt contracts
Cost Structure	Sensor/PCB dev, Cloud infra, App support, Marketing, R&D modules

Market Segment

Business Type	How it Works
B2C	Sell to consumers via e-commerce. Track vitals, get alerts, gamification, premium subscriptions
B2B	Hospitals use for patient monitoring. Licenses and SaaS for dashboards
B2G	Govt deploys in rural PHCs. Monitored via dashboards. Alerts to health workers
B2B2C	Through pharmacies (NFC vials) or insurers bundling HealthBand

A. Gamified Leaderboard + Free Checkups

- Users earn streaks for daily health logs
- Top scorers get monthly free checkups at partnered hospitals
- Hospitals upsell services during visits
- Revenue: Subscriptions + hospital lead commissions

B. NFC Vial Ecosystem

- Each med vial gets an NFC tag from partner pharmacy
- HealthBand links dose, reminders, refills to the tag
- App alerts user + notifies pharmacy for refills
- Revenue: Commission per vial + upsell smart kits

Development Lifecycle

